

SAN FRANCISCO HARBOR, CA, USA

WMO: 994016

Lat: 37.806N

Lon: 122.466W

Elev: 2

StdP: 101.30

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 5.3	(c) 6.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.8	(k) 12.1	(l) 9.0	(m) 11.6	(n) 3.0	(o) 200	(p) 0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 4.0	(c) 22.2	(d) N/A	(e) 20.2	(f) N/A	(g) 18.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.3	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 8.1	(c) 7.3	(e) 3.9	(f) 27.5	(g) 1.1	(h) 2.0	(i) 3.1	(j) 28.9	(k) 2.5	(l) 30.1	(m) 1.8	(n) 31.3	(o) 1.0	(p) 32.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	10.8	11.0	11.4	11.8	12.5	13.5	14.0	14.7	15.1	15.1	13.0	10.7
	DBStd	2.55	2.10	2.32	1.93	1.78	1.74	1.59	1.46	1.64	2.07	2.40	2.22	2.21
	HDD10.0	59	16	14	6	2	0	0	0	0	0	0	2	18
	HDD18.3	2030	233	206	214	197	182	147	133	114	99	106	161	238
	CDD10.0	1086	41	42	51	57	78	104	125	145	154	158	92	39
	CDD18.3	13	0	0	0	1	1	1	0	1	4	5	1	0
	CDH23.3	39	0	0	0	3	3	5	0	4	11	12	0	0
	CDH26.7	4	0	0	0	0	1	1	0	1	1	1	0	0
Wind	WSAvg	3.4	3.2	2.9	3.1	3.5	3.7	4.2	4.0	3.7	3.3	2.8	2.7	3.0
Precipitation	PrecAvg	521	114	94	73	34	9	3	1	2	4	27	70	89
	PrecMax	974	286	346	222	135	60	22	9	17	32	185	202	273
	PrecMin	249	6	2	1	0	0	0	0	0	0	0	0	0
	PrecStd	174	76	75	54	34	14	5	2	4	7	35	55	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.0	19.4	22.4	22.4	23.7	19.1	21.6	25.1	24.9	21.5	17.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	15.8	16.9	16.5	17.4	17.8	18.1	17.9	19.0	20.7	21.7	19.0	16.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	14.7	15.4	15.2	15.2	15.8	16.5	17.1	17.9	18.9	19.6	17.1	15.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	13.8	14.2	14.2	14.1	14.9	15.6	16.3	17.0	17.8	18.0	15.9	14.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.6	4.6	4.4	3.8	3.7	2.9	3.0	4.0	4.8	4.6	4.2
		MCDBR	5.6	6.3	6.0	6.5	6.2	5.4	3.7	4.3	7.2	8.1	6.5	4.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.323	0.329	0.340	0.357	0.363	0.357	0.367	0.344	0.345	0.337	0.322	0.324	
	taud	2.564	2.545	2.506	2.423	2.436	2.457	2.434	2.499	2.482	2.548	2.569	2.567	
	Ebn at Noon	866	905	925	922	917	919	907	923	906	883	859	838	
	Edh at Noon	75	86	98	113	114	111	113	104	99	85	74	71	
All-Sky Solar Radiation	RadAvg	2.38	3.31	4.67	6.13	7.01	7.64	7.17	6.38	5.44	4.02	2.80	2.08	
	RadStd	0.33	0.41	0.42	0.48	0.47	0.34	0.26	0.30	0.21	0.27	0.19	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-110	+135	N/A

Nomenclature: See separate page